

Guia de Despliegue On-Premise — SIGAI-SES

1. Objetivo

Esta guia describe el despliegue del sistema SIGAI-SES en un servidor corporativo local (on-premise) utilizando Nginx como reverse proxy, MySQL propio y certificados SSL con Let's Encrypt. Esta opcion es recomendada para clientes que requieren control total sobre sus datos y tienen infraestructura propia.

2. Arquitectura del Despliegue On-Premise



3. Requisitos del Servidor

Hardware Recomendado

Recurso	Minimo	Recomendado
CPU	2 cores	4 cores
RAM	4 GB	8 GB
Disco	20 GB SSD	50 GB SSD
Red	100 Mbps	1 Gbps

Software Requerido

Componente	Version
Sistema Operativo	Ubuntu Server 22.04 LTS o superior
Python	3.12+
Node.js	18+ (solo para build)
MySQL	8.0+ o MariaDB 10.5+
Nginx	1.24+
Certbot	Ultima version (Let's Encrypt)
Git	Ultima version

4. Instalacion Paso a Paso

4.1 Preparar el servidor

```
# Actualizar sistema
sudo apt update && sudo apt upgrade -y

# Instalar dependencias del sistema
sudo apt install -y python3.12 python3.12-venv python3-pip \
mysql-server-8.0 nginx certbot python3-certbot-nginx \
git curl wget

# Configurar MySQL
sudo mysql_secure_installation
sudo systemctl enable mysql
sudo systemctl start mysql
```

4.2 Clonar el repositorio

```
cd /opt
sudo git clone https://github.com/TU_USUARIO/proyecto-sigai-ses.git sigai-ses
cd sigai-ses
```

4.3 Configurar Backend

```
cd Backend

# Crear entorno virtual
python3.12 -m venv .venv
source .venv/bin/activate

# Instalar dependencias
```

```
pip install -r requirements.txt

# Configurar variables de entorno
cp .env.example .env
nano .env
```

Configuracion de .env para produccion:

```
DATABASE_URL=mysql+aiomysql://sigai:password_local@localhost:3306/sigai_ses_db
SECRET_KEY=generar_con: openssl rand -hex 32
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=480
CORS_ALLOWED_ORIGINS=https://sigai.securitas.com.co
ADMIN_EMAIL=admin@securitas.com
ADMIN_PASSWORD=Admin123!
BACKEND_PORT=8000
```

4.4 Configurar Base de Datos

```
# Crear base de datos y usuario
sudo mysql -u root -p
```

```
CREATE DATABASE IF NOT EXISTS sigai_ses_db CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
CREATE USER IF NOT EXISTS 'sigai'@'localhost' IDENTIFIED BY 'password_local';
GRANT ALL PRIVILEGES ON sigai_ses_db.* TO 'sigai'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

```
# Ejecutar migraciones
alembic upgrade head

# Verificar
python scripts/init_db.py
```

4.5 Configurar Servicio Systemd (Backend)

Crear archivo `/etc/systemd/system/sigai-backend.service`:

```
[Unit]
Description=SIGAI-SES Backend (FastAPI)
After=network.target mysql.service
```

```
[Service]
Type=simple
User=sigai
Group=sigai
WorkingDirectory=/opt/sigai-ses/Backend
Environment=PATH=/opt/sigai-ses/Backend/.venv/bin:/usr/bin
ExecStart=/opt/sigai-ses/Backend/.venv/bin/uvicorn app.main:app --host 127.0.0.1 -
-port 8000 --workers 4
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
```

```
# Crear usuario del sistema
sudo useradd -r -s /bin/false sigai
sudo chown -R sigai:sigai /opt/sigai-ses

# Habilitar e iniciar servicio
sudo systemctl daemon-reload
sudo systemctl enable sigai-backend
sudo systemctl start sigai-backend
sudo systemctl status sigai-backend
```

4.6 Construir Frontend

```
cd /opt/sigai-ses/Frontend

# Instalar dependencias
npm ci

# Configurar .env
cp .env.example .env
echo "VITE_API_BASE_URL=/api/v1" > .env

# Build de produccion
npm run build

# El build se genera en Frontend/dist/
```

4.7 Configurar Nginx

Crear archivo `/etc/nginx/sites-available/sigai-ses`:

```
server {
    listen 80;
```

```
server_name sigai.securitas.com.co;

# Redireccion a HTTPS
return 301 https://$server_name$request_uri;
}

server {
    listen 443 ssl http2;
    server_name sigai.securitas.com.co;

    # SSL (generar con certbot)
    ssl_certificate /etc/letsencrypt/live/sigai.securitas.com.co/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/sigai.securitas.com.co/privkey.pem;
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers HIGH:!aNULL:!MD5;

    # Security headers
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header Referrer-Policy "strict-origin-when-cross-origin" always;
    add_header Content-Security-Policy "default-src 'self'; script-src 'self';
style-src 'self' 'unsafe-inline'; img-src 'self' data;; font-src 'self'" always;

    # Frontend (build estatico)
    root /opt/sigai-ses/Frontend/dist;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }

    # API Backend
    location /api/ {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 300s;
        proxy_send_timeout 300s;
    }

    # Archivos estaticos del backend (avatars, logos)
    location /static/ {
        alias /opt/sigai-ses/Backend/app/static/;
    }

    # Logs
    access_log /var/log/nginx/sigai-access.log;
    error_log /var/log/nginx/sigai-error.log;
}
```

```
# Habilitar sitio
sudo ln -s /etc/nginx/sites-available/sigai-ses /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx
```

4.8 Configurar SSL con Let's Encrypt

```
# Generar certificado SSL
sudo certbot --nginx -d sigai.securitas.com.co

# Verificar renovacion automatica
sudo certbot renew --dry-run
```

5. Verificacion

```
# Verificar servicios
sudo systemctl status sigai-backend
sudo systemctl status nginx
sudo systemctl status mysql

# Probar endpoints
curl -k https://sigai.securitas.com.co/api/v1/health
curl -k https://sigai.securitas.com.co/api/v1/health/db

# Probar frontend (abrir en navegador)
# https://sigai.securitas.com.co
```

6. Mantenimiento

RespalDOS automaticos (cron diario)

```
sudo crontab -e
# Anadir:
0 2 * * * /opt/sigai-ses/Backend/.venv/bin/python /opt/sigai-
ses/Backend/scripts/backup_db.py
```

Evaluacion de alertas (cron cada 30 min)

```
* /30 * * * * /opt/sigai-ses/Backend/.venv/bin/python /opt/sigai-ses/Backend/scripts/trigger_alerts.py
```

Rotacion de logs

Nginx ya incluye rotacion de logs. Para los logs de Uvicorn, configurar logrotate:

```
sudo nano /etc/logrotate.d/sigai-backend
```

```
/opt/sigai-ses/Backend/logs/*.log {  
    daily  
    rotate 30  
    compress  
    delaycompress  
    missingok  
    notifempty  
    copytruncate  
}
```

Actualizacion del sistema

```
cd /opt/sigai-ses  
git pull origin main  
  
# Backend  
sudo systemctl stop sigai-backend  
cd Backend  
source .venv/bin/activate  
pip install -r requirements.txt  
alembic upgrade head  
sudo systemctl start sigai-backend  
  
# Frontend  
cd ../Frontend  
npm ci  
npm run build  
sudo systemctl reload nginx
```

7. Firewall (Puertos Requeridos)

```
# Configurar UFW  
sudo ufw allow 22/tcp          # SSH
```

```
sudo ufw allow 80/tcp      # HTTP (redireccion)
sudo ufw allow 443/tcp     # HTTPS
sudo ufw allow 3306/tcp    # MySQL (solo si acceso remoto necesario)
sudo ufw enable
sudo ufw status verbose
```

Documento actualizado al: Julio 2026 - v1.0